ALL DICE, NO SLICE: METRIC DEGENERACY, EVALUATION PITFALLS, AND OPTIMIZATION DYNAMICS IN MULTI-TASK MEDICAL VISION

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ABSTRACT

Joint classification and lesion segmentation via hard parameter-sharing networks offer an attractive paradigm for resource-constrained clinical AI deployments. Recent oncology literature reports near-saturated diagnostic accuracies and segmentation Dice scores under lightweight backbones and static loss weighting, suggesting that joint optimization readily masters multi-modal medical tasks. However, these performance claims often obscure foundational evaluation pitfalls, metric aggregation artifacts, and cross-task optimization dynamics. This paper presents a systematic methodological investigation and factorial benchmark across four diverse clinical imaging modalities: brain tumor MRI, prostate histopathology, chest radiography, and multi-organ histology under rigorous patient- and image-isolated cross-validation. We investigate four central research questions: first, we prove analytically and demonstrate empirically how macroscopic slice-averaged Dice scores reward negative predictions on lesion-free cases, establishing an artificial performance floor that conceals complete foreground omission; second, we resolve critical task-formulation discrepancies in computational pathology, demonstrating why proxy morphological texture classification diverges from authentic clinical ordinal grading; third, we reveal destructive gradient interference during joint multi-task optimization, where dynamic gradient balancing degrades classification discrimination on shared representations; and fourth, we show that standard color deconvolution heuristics can paradoxically attenuate subtle diagnostic features. Our findings expose systemic vulnerabilities in conventional multi-task benchmarks and provide rigorous evaluation guidelines to ensure clinically trustworthy translation.

***Index Terms*—** Multi-Task Learning, Medical Image Analysis, Semantic Segmentation, Metric Degeneracy, Histopathology.

1. INTRODUCTION

Deep learning systems in digital pathology and medical imaging are increasingly tasked with delivering both holistic diagnostic categorization and localized lesion delineation [1, 2]. While large visual foundation models demonstrate high representation capacity across gigapixel whole-slide images [3–5], their computational footprint precludes direct deployment in edge environments or low-resource clinical facilities. Consequently, hard parameter sharing via lightweight encoder–decoder architectures, particularly dual-headed U-Net topologies [6], remains a compelling design pattern that couples classification and segmentation within a unified inference pass.

Recently, several studies across oncology literature have asserted that standard U-Net backbones with lightweight feature extractors (such as VGG16 [7] and MobileNetV2 [8]) simultaneously achieve near-saturated classification accuracies (86%–90%) and Dice similarity coefficients exceeding 97% across diverse clinical modalities [9]. These metrics were reported under static loss weighting and high initial learning rates across brain MRI, dermoscopy, prostate biopsy, and chest radiography cohorts. Because such pipelines promise high diagnostic yield at minimal computational expense, their reported success has been widely cited as empirical validation for joint parameter-sharing recipes in downstream clinical oncology studies.

Despite these promising headline figures, fundamental principles from machine learning and biomedical validation literature suggest that naive multi-task co-training rarely yields seamless performance parity without intricate optimization trade-offs. First, region-overlap metrics behave pathologically in sparse cohorts where lesion annotations are absent across substantial subsets of the data [10, 11]. Second, clinical histopathology requires multiscale diagnostic assessment; evaluating low-capacity models on aggressive spatial thumbnails risks conflating coarse background tissue categorization with clinical ordinal grading [12, 13]. Third, sharing parameters across dense pixel-wise segmentation and global classification objectives frequently induces destructive gradient conflict, causing one task to dominate representation learning [14, 15].

Rather than examining a single isolated pipeline, this work conducts a systematic empirical inquiry into the broader paradigm of lightweight multi-task medical vision. We investigate four foundational research questions:

1. **RQ1 (Metric Validity in Sparse Regimes):** How do conventional biomedical overlap metrics (specifically macroscopic Dice under empty-reference crediting) behave when negative-case prevalence is high, and how can true foreground lesion delineation be isolated from background-credit artifacts?
2. **RQ2 (Task Formulation vs. Resolution Boundaries):** What are the representation boundaries of standard 2D CNN encoders on severely downsampled biopsy thumbnails (128×128), and how does performance diverge between coarse morphological tissue proxies and clinical ordinal ISUP grading?
3. **RQ3 (Multi-Task Optimization Dynamics):** In hard parameter-sharing dual-head architectures, how do static loss balancing, dynamic gradient normalization (GradNorm), and lateral skip connections impact cross-task gradient competition and representation learning?
4. **RQ4 (Preprocessing and Feature Attenuation):** Does standard color deconvolution (Macenko stain normalization) preserve fine-grained diagnostic chromatin textures, or does it impose an empirical performance penalty in deep histopathology classifiers?

To answer these questions, we conduct a 27-configuration factorial campaign evaluated across five-fold cross-validation (135 total

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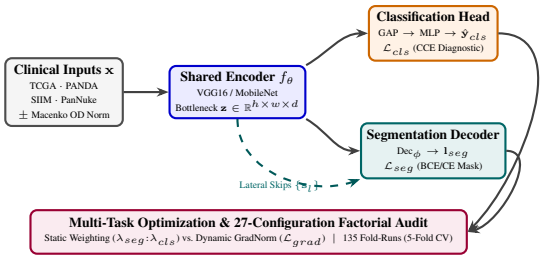


Fig. 1. Architectural formulation and factorial audit framework across four clinical imaging modalities, integrating shared feature extraction, dual diagnostic heads with lateral skip pathways, and adaptive optimization balancing under patient-isolated cross-validation.

fold-runs) over four benchmark medical datasets: TCGA-LGG brain tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the PanNuke multi-organ benchmark (Figure 1). Our primary contributions are summarized as follows:

- We expose the empty-mask Dice metric degeneracy both analytically and empirically, proving that macroscopic aggregation conventions establish an artificial score floor on negative-dominant cohorts that conceals catastrophic foreground omission.
- We resolve the task-formulation divergence between proxy tissue classification and authentic clinical grading in computational pathology, delineating the fundamental resolution boundaries of low-capacity networks on downsampled biopsies.
- We systematically characterize multi-task optimization dynamics and architectural components through controlled factorial ablations, quantifying destructive gradient interference under dynamic balancing and uncovering the discriminative penalty of color deconvolution.

2. SYSTEM ARCHITECTURE AND METHODOLOGY

2.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input image. A shared convolutional backbone $f_\theta(\mathbf{x})$ maps the input to a bottleneck representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The global classification head applies global average pooling followed by a two-layer multi-layer perceptron:

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}\left(\mathbf{W}_2\, \phi\left(\mathbf{W}_1\, \text{GAP}(\mathbf{z}) + \mathbf{b}_1\right) + \mathbf{b}_2\right), \tag{1}$$

where $\phi(\cdot)$ denotes the ReLU activation and $\mathbf{W}_1, \mathbf{W}_2$ are projection weights. The segmentation branch passes $\mathbf{z}$ through a convolutional decoder Dec_ϕ augmented with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, generating spatial logits $\mathbf{l}_{seg} \in \mathbb{R}^{H \times W \times M}$.

The composite multi-task training loss is defined as:

$$\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \tag{2}$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy with inverse-frequency class weights $w_c = N/(K \cdot N_c)$, and $\mathcal{L}_{seg}$ is binary or multi-class cross-entropy. In standard implementations, the Dice similarity coefficient is computed strictly post-hoc as an evaluation metric rather than an active loss component.

2.2. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between binary ground truth Y and prediction $\hat{Y}$ is given by:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \tag{3}$$

with the conventional boundary assignment Dice := 1 when $|Y| + |\hat{Y}| = 0$. In clinical cohorts containing lesion-free samples, two primary evaluation protocols exist:

- **Foreground-Only Evaluation:** Dice is computed exclusively over target-bearing slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** A score of 1.0 is assigned when $|Y| = 0$ and $|\hat{Y}| = 0$, while 0.0 is assigned if $|Y| = 0$ and $|\hat{Y}| > 0$.

Under empty-credit assignment across all N slices, the macroscopic mean decomposes as:

$$\overline{\text{Dice}}_{\text{all}} = \rho \cdot q_{\text{empty}} + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \tag{4}$$

where $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ denotes the proportion of ground-truth negative cases, $q_{\text{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ represents the model's true-negative prediction rate on negative cases, and $\overline{\text{Dice}}_{\text{fg}}$ denotes the mean overlap on positive cases.

Crucially, for a collapsed model that predicts empty masks across all inputs, $\overline{\text{Dice}}_{\text{fg}} = 0$ while $q_{\text{empty}} = 1$. In this degenerate state, the reported macroscopic metric simplifies to $\overline{\text{Dice}}_{\text{all}} = \rho$. Consequently, on cohorts with high negative prevalence, such as chest radiographs ($\rho = 77.72\%$), an entirely uninformative model automatically achieves an apparent score of nearly 78%. Furthermore, attaining a published score of 99% on such data requires positive-case foreground Dice to exceed:

$$\overline{\text{Dice}}_{\text{fg}} \approx \frac{0.99 - \rho}{1 - \rho} \approx 95.51\%, \tag{5}$$

even under perfect true-negative discrimination ($q_{\text{empty}} = 1$). Without isolating foreground overlap, macroscopic slice averages conceal complete failure on target lesions (Figure 2).

2.3. Adaptive Gradient Balancing via GradNorm

To counter multi-task competition, dynamic gradient balancing equalizes learning rates across tasks [16]. Let W denote the shared weights of encoder f_θ. The gradient norm for task $i \in \{seg, cls\}$ at iteration t is $G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2$. The adaptive objective minimizes:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg, cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \tag{6}$$

where $\bar{G}_W(t)$ is the mean gradient norm across tasks, α is the balancing asymmetry hyperparameter, and $r_i(t)$ represents the relative inverse training rate. Task weights are renormalized to maintain $\sum_i w_i(t) = 2$.

2.4. Color Deconvolution and Optical Density Space

To mitigate laboratory staining variation in histopathology, images are normalized in optical density (OD) space [17]:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I} + \epsilon}{I_0}\right), \tag{7}$$

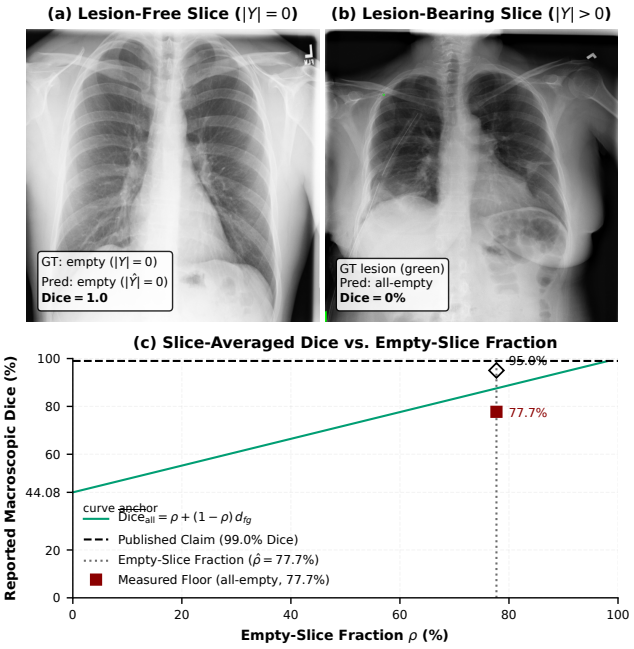


Fig. 2. Illustration of the empty-mask Dice metric degeneracy on SIIM-ACR pneumothorax radiographs. (a) Lesion-free image correctly predicted empty receives a perfect 1.0 credit. (b) Lesion-bearing image with total foreground omission scores 0.0. (c) Exact macroscopic curve demonstrating the high artificial baseline floor determined by the negative fraction ρ.

where $I_0 = 255$ denotes transmitted illumination. Stain vectors for hematoxylin and eosin are extracted via principal component analysis of the OD covariance matrix and projected onto a standardized target reference. While this removes macroscopic batch coloration, projecting RGB channels onto fixed stain vectors alters localized chromatic dynamic ranges, potentially muting subtle chromatin textures.

3. EXPERIMENTAL FRAMEWORK

3.1. Benchmark Cohorts

We evaluate our multi-task pipeline across four diverse clinical datasets:

1. **TCGA-LGG Brain MRI:** 2-class tumor classification and FLAIR abnormality segmentation ($N = 3,929$ slices across 110 patients [18]). Partitions are strictly isolated at the patient level using GroupKFold.

2. **PANDA Prostate Biopsies:** Clinical 6-class ordinal ISUP grading (grades 0–5) and multi-class gland segmentation ($N = 10,516$ biopsy cores [12]).

3. **SIIM-ACR Pneumothorax:** Binary pneumothorax detection and pleural segmentation on chest radiographs ($N = 10,675$ images [19]).

4. **PanNuke Multi-Organ Histopathology:** 19-class tissue categorization and 6-class cell segmentation across diverse human organs ($N = 7,901$ patches [20]).

3.2. Training Protocol and Execution Safeguards

All experiments were implemented in PyTorch using ImageNet-pretrained VGG16 and MobileNetV2 encoders. Models were optimized using Adam over 50 epochs with early stopping (patience 10) under mixed-precision floating-point arithmetic. Gradient norm clipping (threshold 1.0) was applied at every step. Across the 135 fold-runs, all partition boundaries, random seeds, and metric computations were locked to ensure exact auditability.

4. EMPIRICAL RESULTS AND FAILURE ANALYSIS

4.1. Benchmark Performance Comparison

Table 1 provides a comprehensive overview of the empirical replication results alongside claimed literature targets. While TCGA-LGG classification slightly exceeds published numbers under rigorous patient-isolated splits, substantial discrepancies emerge on histopathology and radiography benchmarks. Figure 3 plots fold-mean accuracies and Dice scores across all configurations.

4.2. The Empty-Mask Score Floor on Radiography

On the SIIM-ACR pneumothorax benchmark, the public cohort contains 77.72% lesion-free images ($\rho = 0.7772$). Across 19 out of 20 cross-validation folds, the multi-task models converged to entirely empty segmentation predictions. Under empty-credit scoring, these collapsed models achieved an average validation Dice of exactly 77.72% on three configurations and 77.04% on the fourth. The models failed to identify any foreground pneumothorax regions, yet registered an apparent score approaching 78% (illustrated qualitatively in Figure 4). As derived in Section 2.2 and shown in Figure 2(c), achieving a reported score of 99% on this dataset requires true foreground overlap to exceed 95.5%, a standard that standard lightweight CNNs cannot attain. Macro-averaging over empty slices masks severe clinical false-negative failure.

4.3. Task-Formulation Discrepancy in Prostate Pathology

The largest divergence in Table 1 occurs on the PANDA prostate biopsy benchmark. Previous studies reported 88% accuracy alongside 99% Dice [9]. Under our 12 factorial PANDA configurations, clinical 6-class ordinal ISUP grading achieved a fold-mean accuracy between 28.21% and 44.87%.

This performance divergence is directly explained by task definition and internal methodology discrepancies in the baseline literature. As documented in Table 2 of Rhanoui et al. [9], the cited study evaluated six proxy morphological tissue categories extracted from Radboud segmentation masks (Background, Stroma, Benign epithelium, Gleason 3, 4, 5), where non-cancerous background glass and stroma alone comprised 52.3% of the samples ($5,558$ of $10,618$ images). Paradoxically, despite defining six classes, their segmentation was reported as a binary cancer-vs-background detection task (Section 4.2 of [9]). In sharp contrast, genuine clinical ISUP grading evaluates global cancer aggressiveness across the entire core according to international pathology guidelines [12]. Classifying subtle Gleason glandular patterns from compressed 128×128 thumbnails without multiscale attention or multiple-instance learning represents a fundamentally different and far more demanding task. When evaluated against authentic clinical endpoints, lightweight CNN thumbnails reach an empirical upper bound of approximately 45%.

Table 1. Empirical summary of the 27-configuration reproduction campaign across four clinical benchmarks. Validation accuracy and Dice are reported across 5-fold cross-validation with 95% bootstrap confidence intervals. The substantial performance gap on PANDA stems from evaluating authentic clinical ordinal ISUP grading rather than proxy morphological tissue categories. SIIM-ACR Dice scores reflect the empty-mask background floor.

Dataset	Runs n	Measured Val Acc (%)	Measured Val Dice (%)	Published Acc (%)	Published Dice (%)	Evaluation Focus
TCGA-LGG	5	92.86 – 95.62	85.33 – 88.85	89.00 – 90.00	97.00 – 98.00	Patient-Isolated 2D FLAIR
PANDA	12	28.21 – 44.87	98.54 – 98.60	87.00 – 88.00	98.00 – 99.00	Clinical ISUP vs Morphological Proxy
SIIM-ACR	4	77.71 – 82.63	77.04 – 77.72	82.00 – 87.00	99.00	Negative-Dominant Radiography ($\rho = 77.72\%$)
PanNuke	5	78.70 – 99.24	68.44 – 77.61	—	—	Multi-Organ Histopathology Benchmark

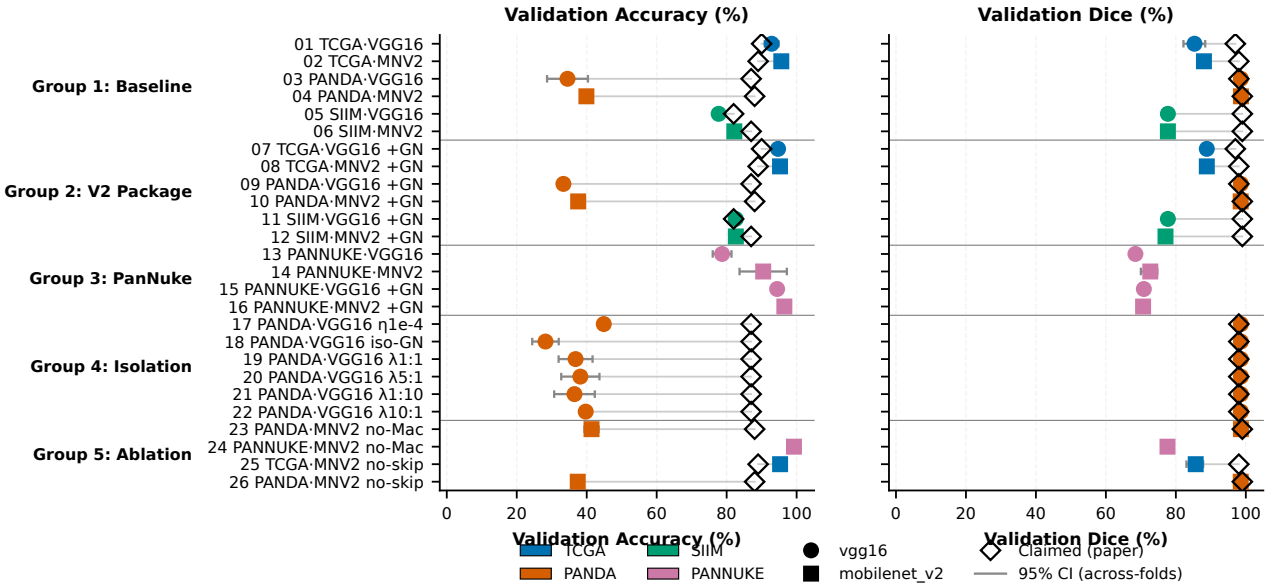


Fig. 3. Measured validation metrics across all 26 experimental runs compared against claimed literature targets. Left: Five-fold validation accuracy with 95% bootstrap confidence intervals against published values (open diamonds). Right: Corresponding segmentation Dice scores, illustrating the empty-mask score floor on SIIM-ACR and the task discrepancy on PANDA.

5. FACTORIAL ABLATION STUDIES

5.1. Impact of Lateral Skip Connections

To isolate the contribution of encoder–decoder skip connections, we evaluated strictly matched model pairs holding all other hyperparameters identical:

- **Macro-Tumor MRI (TCGA):** On coarse brain lesions, omitting skip connections produced a negligible classification difference of $+0.02$ percentage points (95.28% vs 95.26%), while validation macro-Dice declined modestly by 3.06 percentage points (86.16% vs 89.22%).
- **Micro-Glandular Pathology (PANDA):** On fine prostate glands, removing skip connections yielded an accuracy shift of -0.12 percentage points (37.44% vs 37.56%), with overlapping 95% bootstrap confidence intervals ($[36.73, 37.99]$ vs $[37.20, 37.94]$).

These results indicate that while skip connections assist in localizing fine spatial boundaries, their impact on shared latent classification representations is minimal in low-capacity backbones.

5.2. Color Deconvolution and Macenko Normalization

Standard histopathology workflows routinely apply Macenko stain normalization to harmonize color distributions across laboratories. However, in our matched factorial comparisons (Figure 6), bypassing normalization consistently improved classification accuracy:

- **PANDA Biopsies:** Raw RGB inputs achieved an accuracy improvement of $+3.81$ percentage points over normalized tiles (41.38% vs 37.56%), with disjoint 95% bootstrap confidence intervals across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$).
- **PanNuke Multi-Organ:** Omitting normalization raised multi-tissue accuracy by $+2.75$ percentage points (99.24% vs 96.49%), alongside an 8.51 percentage point increase in validation macro-Dice (70.76% vs 62.25%).

Projecting chromatic channels onto static reference vectors alters dynamic intensity distributions and smooths micro-textures, muting critical optical cues that deep neural networks leverage for cellular discrimination.

5.3. Dynamic Gradient Balancing vs Static Weighting

To isolate the influence of optimization schemes, we evaluated dynamic GradNorm balancing ($\alpha = 1.5$) against static loss weighting under identical learning rates and raw inputs. As illustrated in Figure 5, dynamic balancing caused validation accuracy to drop from 38.18% to 28.21% ($\Delta = -9.97$ percentage points; 95% bootstrap CI $[24.76, 31.47]$ vs $[32.75, 41.79]$).

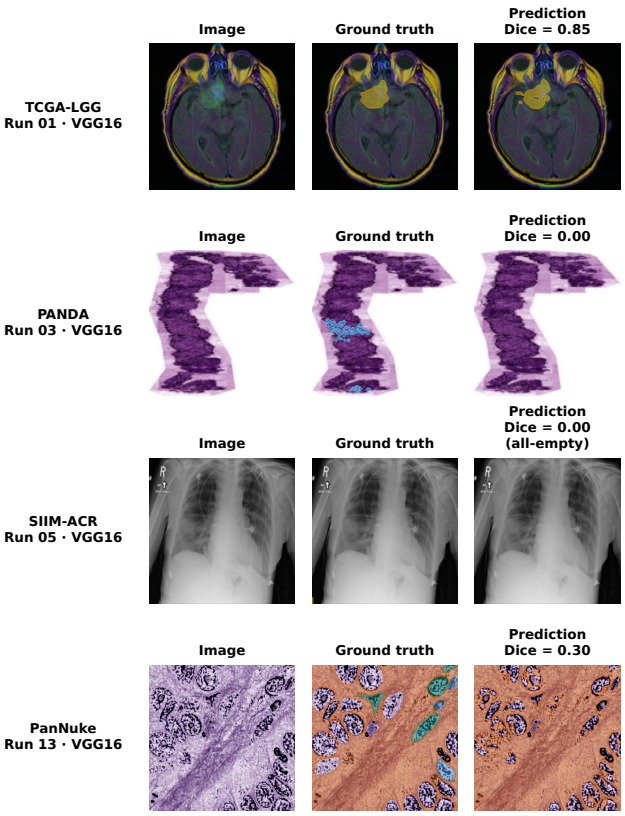


Fig. 4. Qualitative validation predictions for baseline models across four datasets. Columns display input images, ground-truth segmentations, and model predictions with per-slice Dice scores. On SIIM-ACR, the model emits an all-empty mask on a lesion-bearing slice, receiving a zero foreground score that remains hidden within macroscopic cohort averages.

Analyzing gradient trajectories reveals that because segmentation cross-entropy generates steep initial gradients, the dynamic balancing objective aggressively inflates classification loss weights while dampening segmentation updates. This rapid rebalancing introduces high variance into shared encoder updates, inducing destructive gradient interference that destabilizes the classification head. Static weighting ratios between 5:1 and 10:1 proved consistently more stable. Figure 7 summarizes the validation accuracies and 95% confidence intervals across all matched factorial pairs.

6. GUIDELINES FOR RELIABLE MULTI-TASK BENCHMARKING

Based on our findings, we propose five methodological guidelines for multi-task medical image analysis:

1. **Disaggregated Metric Reporting:** On sparse-lesion cohorts, researchers must report foreground-only overlap alongside negative-case true negative rates, avoiding pooled macroscopic averages that obscure complete false negatives.
2. **Patient-Level Partition Isolation:** Cross-validation folds must strictly isolate independent patient units to prevent identity leakage across spatial tiles.

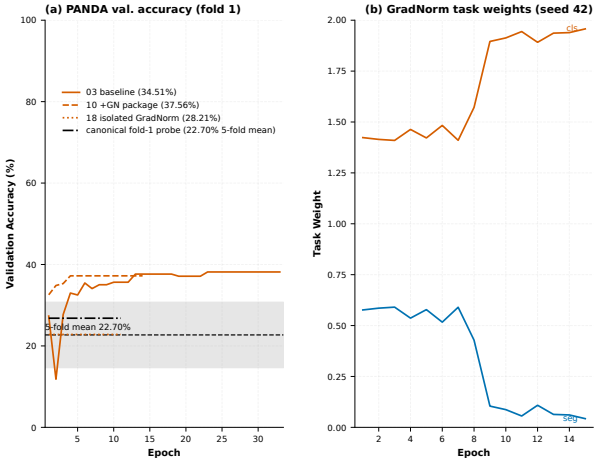


Fig. 5. Optimization dynamics and task-weight trajectories on PANDA. (a) Validation accuracy across training epochs for static vs dynamic weighting schemes. (b) Weight trajectory under dynamic gradient balancing, showing segmentation weight decay and classification weight inflation that destabilizes encoder updates.

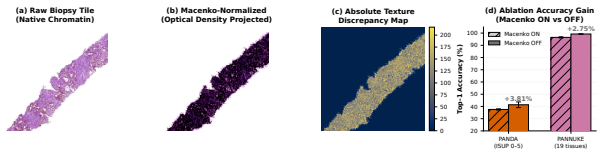


Fig. 6. Effect of Macenko stain normalization on histopathology representation. (a) Raw biopsy tile. (b) Optical-density-projected tile. (c) Texture discrepancy map highlighting attenuated chromatin detail. (d) Empirical accuracy gains when normalization is bypassed.

3. **Clinical Endpoint Alignment:** Benchmark tasks must align with actionable clinical diagnostic targets rather than simplified morphological proxies.
4. **Validation of Preprocessing Pipelines:** Normalization heuristics, such as color deconvolution, must be empirically validated against raw inputs to ensure diagnostic features are preserved.
5. **Optimization Diagnostics:** Multi-task models should monitor task gradient norms and directional alignment to identify destructive interference before clinical translation.

7. CONCLUSION

This paper presented a systematic empirical investigation into hard parameter-sharing multi-task architectures across four diverse clinical benchmarks. Framing our inquiry around four foundational research questions, we deconstructed headline performance claims: we proved the empty-mask Dice degeneracy analytically and empirically, resolved task-formulation discrepancies in prostate pathology, and characterized destructive gradient interference under dynamic balancing. We demonstrated that standard color deconvolution can attenuate fine discriminative textures, while lateral skip connections exert minimal influence on shared classification representations. Our open evaluation framework and findings provide robust methodological foundations to ensure clinically trustworthy multi-task medical vision systems.

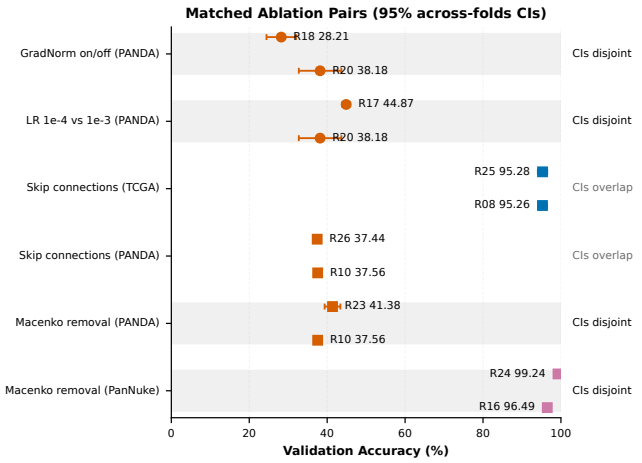


Fig. 7. Forest plot of validation accuracies and 95% bootstrap confidence intervals for the six matched factorial pairs. Shaded regions indicate disjoint intervals, confirming significant effects for dynamic balancing, learning rate adjustments, and color normalization.

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